LEVEL 2 – BASIC MCQs (FORMULAS & CONCEPTS)

1. Basics of Electrochemistry

1. Which of the following is a good ionic conductor?
   a) NaCl solid
   b) NaCl solution
   c) Glass
   d) Diamond
2. Which of these is an insulator?
   a) Copper
   b) NaOH solution
   c) Diamond
   d) NaCl solution
3. Ionic conductivity depends on:
   a) Temperature
   b) Concentration of ions
   c) Mobility of ions
   d) All of the above
4. Molar conductivity of a strong electrolyte:
   a) Increases with dilution
   b) Decreases with dilution
   c) Remains constant
   d) Zero
5. Limiting molar conductivity is:
   a) Conductivity at infinite dilution
   b) Conductivity at zero temperature
   c) Conductivity at standard conditions
   d) Conductivity at high concentration
6. Conductivity of a solution decreases with:
   a) Increase in concentration
   b) Increase in temperature
   c) Increase in ion mobility
   d) None of the above
7. The unit of conductivity is:
   a) $S \cdot m^{-1}$

b) $\Omega \cdot m$

c) $S \cdot cm$

d) $\Omega^{-1} \cdot m^{-2}$

8. Which solution has the highest conductivity?

   a) 0.1 M NaCl

   b) 0.01 M NaCl

   c) 1 M NaCl

   d) 0.1 M sugar solution

9. Which is a weak electrolyte?

   a) NaCl

   b) HCl

   c) CH_3COOH

   d) NaOH

10. Strong electrolytes:

    a) Completely ionize in solution

    b) Partially ionize

    c) Do not ionize

    d) Only conduct at high temperature

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2. Electrolytic and Galvanic Cells

11. In an electrolytic cell, the cathode is:

    a) Positive

    b) Negative

    c) Neutral

    d) None

12. In a galvanic cell, the anode is:

    a) Positive

    b) Negative

    c) Neutral

    d) Depends on solution

13. Which of the following is a galvanic cell?

    a) Electrolysis of NaCl

    b) Zn-Cu Daniell cell

    c) Electroplating of Ag

    d) Electrolysis of water

14. Which electrode undergoes oxidation in a Zn-Cu cell?
    a) Zn
    b) Cu
    c) Both
    d) None
15. The EMF of a galvanic cell is:
    a) Negative
    b) Positive
    c) Zero
    d) Depends on concentration
16. Standard cell notation for Zn-Cu cell is:
    a) Zn | Zn^{2+} || Cu^{2+} | Cu
    b) Cu | Cu^{2+} || Zn^{2+} | Zn
    c) Zn || Cu
    d) Zn | Cu
17. Electrolytic cells require:
    a) External voltage
    b) Spontaneous reaction
    c) High temperature
    d) Catalyst
18. Galvanic cells convert:
    a) Electrical energy → Chemical energy
    b) Chemical energy → Electrical energy
    c) Heat → Electrical energy
    d) None of the above
19. Which statement is correct?
    a) Anode is always positive
    b) Cathode is always negative
    c) Anode is site of oxidation
    d) Cathode is site of oxidation
20. Example of electrolytic cell:
    a) Daniell cell
    b) Zn-Cu cell
    c) Electrolysis of molten NaCl
    d) Lead-acid battery

3. Electrode Potential

21. Standard electrode potential (E°) is measured against:
    a) Ag/AgCl
    b) SHE
    c) Calomel electrode
    d) Any electrode
22. SHE stands for:
    a) Standard Hydrogen Electrode
    b) Saturated Hydrogen Electrode
    c) Standard Halogen Electrode
    d) Saturated Halogen Electrode
23. Nernst equation relates:
    a) Cell potential and concentration
    b) Temperature and resistance
    c) Current and voltage
    d) Charge and mass
24. Cell potential decreases when:
    a) Reactant concentration decreases
    b) Product concentration increases
    c) Both a and b
    d) None
25. Electrode with higher E°:
    a) Stronger oxidizing agent
    b) Stronger reducing agent
    c) Neutral
    d) None
26. For the reaction $Zn^{2+} + 2e^- \rightarrow Zn$, if $[Zn^{2+}] = 0.01$ M, the electrode potential:
    a) Increases
    b) Decreases
    c) Remains same
    d) Zero
27. Reference electrode in potentiometry:
    a) Measures potential
    b) Provides constant potential
    c) Generates current
    d) Conducts electrons

28. E°cell = E°cathode − E°anode
    a) True
    b) False
29. Which is a reduction half-cell?
    a) $Cu^{2+} + 2e^- \rightarrow Cu$
    b) $Zn \rightarrow Zn^{2+} + 2e^-$
    c) Both
    d) None
30. Nernst equation at 25°C:
    a) $E = E° - 0.059/n \log Q$
    b) $E = E° + 0.059/n \log Q$
    c) $E = E° - 0.059n \log Q$
    d) $E = E° + 0.059n \log Q$

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4. Electrochemical Series

31. A metal with lower E°:
    a) Corrodes faster
    b) Corrodes slower
    c) Cannot corrode
    d) None
32. Zn or Cu, which will displace Cu^{2+}?
    a) Zn
    b) Cu
    c) Both
    d) None
33. Electrochemical series helps to:
    a) Predict direction of reaction
    b) Predict corrosion tendency
    c) Identify oxidizing/reducing agents
    d) All of the above
34. Which metal is least reactive?
    a) K
    b) Mg
    c) Au
    d) Zn

35. Strongest reducing agent among:
    a) Li
    b) Cu
    c) Ag
    d) Au
36. Most easily oxidized metal:
    a) Na
    b) Al
    c) Fe
    d) Cu
37. Corrosion tendency decreases:
    a) Moving down the series
    b) Moving up the series
    c) Random
    d) None
38. Noble metals are:
    a) Highly reactive
    b) Less reactive
    c) Oxidizing agents
    d) None
39. Reaction $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$:
    a) Spontaneous
    b) Non-spontaneous
    c) Equilibrium
    d) Cannot occur
40. Oxidation occurs at:
    a) Anode
    b) Cathode
    c) Both
    d) None

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5. Faraday's Laws of Electrolysis

41. Mass of substance deposited is directly proportional to:
    a) Quantity of electricity
    b) Time only

c) Current only

d) Voltage

42. Electrochemical equivalent is:

a) Mass deposited per unit charge

b) Mass deposited per mole

c) Charge per mole

d) None

43. Mass of substance deposited = $Z \times Q$

a) True

b) False

44. $Q = I \times t$

a) True

b) False

45. Two Faradays deposit:

a) 1 mole of electrons

b) 2 moles of electrons

c) 1 mole of substance

d) 2 moles of substance

46. Example of electroplating:

a) Cu plating

b) Electrolysis of water

c) Daniell cell

d) None

47. Electrorefining is used to:

a) Purify metals

b) Corrode metals

c) Generate electricity

d) Electrolyze water

48. Water electrolysis produces:

a) H_2 at cathode, O_2 at anode

b) O_2 at cathode, H_2 at anode

c) H_2 and O_2 at cathode

d) Only H_2

49. 1 mole of electrons =

a) 96500 C

b) 1 F

c) Both a and b
d) None

50. Mass deposited ∝ Equivalent weight of substance
a) True
b) False

I've drafted **50 MCQs here for Level 2**, covering the **first 5 topics**. To keep it organized, the next sets will cover:

- **6. Types of Electrochemical Cells**
- **7. Corrosion**
- **8. Batteries**
- **9. Conductometry, Potentiometry, Voltammetry**
- **10. Applications**

If you want, I can **continue and complete all 150 MCQs for Level 2** next, topic-wise.

Do you want me to continue?

yes

Perfect! Let's continue **Level 2 – Basic MCQs**, covering the remaining topics (6–10) to complete **~150 questions**.

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6. Types of Electrochemical Cells

51. Which is a primary cell?
a) Dry cell
b) Lead-acid battery
c) Li-ion battery
d) Ni-Cd battery

52. Which is a secondary/rechargeable cell?
a) Leclanché cell
b) Lead-acid battery

c) Dry cell

d) Alkaline cell

53. Which fuel cell uses H_2 and O_2?

a) Hydrogen-oxygen fuel cell

b) Alkaline fuel cell

c) Methanol fuel cell

d) Lead-acid cell

54. Advantage of secondary cell:

a) Rechargeable

b) Non-rechargeable

c) Generates heat

d) Single-use

55. Disadvantage of primary cell:

a) Cannot be recharged

b) Expensive

c) Low energy density

d) None

56. Ni-Cd battery reaction involves:

a) Nickel hydroxide and cadmium

b) Lead and sulfuric acid

c) Lithium and cobalt oxide

d) Zinc and manganese dioxide

57. Lead-acid battery contains:

a) Pb and PbO_2 electrodes, H_2SO_4 electrolyte

b) Zn and Cu electrodes, NaOH electrolyte

c) Li and C electrodes, organic electrolyte

d) None

58. Hydrogen-oxygen fuel cell produces:

a) H_2O + electricity

b) $H_2 + O_2$

c) Heat only

d) O_2 only

59. Li-ion battery is preferred for:

a) Mobile phones

b) Flashlights

c) Dry cells

d) Car battery

60. Which is a non-rechargeable battery?

a) Dry cell

b) Ni-Cd

c) Li-ion

d) Lead-acid

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7. Corrosion

61. Rusting is an example of:

a) Electrochemical corrosion

b) Chemical corrosion

c) Physical corrosion

d) None

62. Galvanization prevents corrosion by:

a) Coating with Zn

b) Coating with Cu

c) Alloying with Fe

d) Electrolysis

63. Cathodic protection works by:

a) Making metal cathode

b) Making metal anode

c) Removing water

d) Painting

64. Alloying prevents corrosion because:

a) Forms protective oxide layer

b) Changes color

c) Increases solubility

d) None

65. Electrochemical corrosion occurs when:

a) Two metals are in contact in an electrolyte

b) Metal is dry

c) Metal is insulated

d) No reaction

66. Which metal corrodes faster in water?
    a) Fe
    b) Cu
    c) Au
    d) Ag
67. Rust formation requires:
    a) Oxygen and water
    b) Water only
    c) Oxygen only
    d) Salt only
68. Sacrificial anode protects:
    a) Steel structure
    b) Zinc coating
    c) Copper wires
    d) Aluminum foil
69. Anode corrodes in:
    a) Galvanic corrosion
    b) Cathodic protection
    c) Painting
    d) None
70. Which solution accelerates rusting?
    a) NaCl solution
    b) Pure water
    c) Alcohol
    d) Oil

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8. Batteries and Storage Devices

71. Primary battery provides energy by:
    a) Spontaneous chemical reaction
    b) Rechargeable reaction
    c) Electrolysis
    d) None
72. Secondary batteries can be:
    a) Recharged
    b) Non-rechargeable

c) Dry cells only

d) Fuel cells

73. Lead-acid battery anode reaction:

a) $Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^-$

b) $PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$

c) Both

d) None

74. Ni-Cd battery has:

a) High recharge cycles

b) Limited cycles

c) Only primary use

d) None

75. Li-ion battery cathode:

a) $LiCoO_2$

b) PbO_2

c) Cd

d) Zn

76. Battery efficiency depends on:

a) Internal resistance

b) Temperature

c) Charge/discharge rate

d) All of the above

77. Application of primary batteries:

a) Remote controls

b) Cars

c) UPS systems

d) Grid storage

78. Application of secondary batteries:

a) Mobile phones

b) Single-use devices

c) Flashlights

d) None

79. Advantage of Li-ion:

a) High energy density

b) Low recharge cycles

c) Heavy weight

d) Limited application

80. Disadvantage of lead-acid:

a) Heavy

b) Non-rechargeable

c) Expensive

d) Low voltage

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9. Conductometric, Potentiometric, and Voltammetric Methods

81. Conductometry measures:

a) Conductivity change

b) Cell potential

c) Current-voltage relation

d) Mass deposition

82. Potentiometry measures:

a) Cell potential

b) Conductivity

c) Ion mobility

d) Charge

83. Voltammetry is used for:

a) Analytical applications

b) Battery testing

c) Electroplating

d) Electrorefining

84. Conductivity increases with:

a) Temperature

b) Solvent viscosity increase

c) Concentration decrease for strong electrolytes

d) None

85. Which is a reference electrode in potentiometry?

a) Ag/AgCl

b) Cu

c) Zn

d) Fe

86. Advantage of voltammetry:
    a) Detect trace metals
    b) Measure voltage only
    c) Only qualitative
    d) None
87. Conductometric titration can detect:
    a) End point of acid-base reactions
    b) Metal deposition
    c) Fuel cell efficiency
    d) Corrosion rate
88. Potentiometric titration uses:
    a) Voltage measurement
    b) Current measurement
    c) Resistance measurement
    d) Mass measurement
89. Limiting molar conductivity is measured at:
    a) Infinite dilution
    b) Standard conditions
    c) High concentration
    d) Zero temperature
90. Conductivity unit:
    a) $S \cdot m^{-1}$
    b) $\Omega \cdot m$
    c) $V \cdot A^{-1}$
    d) $C \cdot s^{-1}$

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10. Applications of Electrochemistry

91. Electroplating is used to:
    a) Coat metals
    b) Purify metals
    c) Generate electricity
    d) Measure conductivity
92. Electrorefining purifies:
    a) Copper
    b) Iron

c) Aluminum

d) Sodium

93. Chlor-alkali process produces:

a) Cl_2 and NaOH

b) HCl only

c) NaCl only

d) H_2O only

94. Electrolysis of molten salts is used to:

a) Extract metals

b) Generate heat

c) Measure conductivity

d) None

95. Conductometry determines:

a) Concentration of ions

b) Cell EMF

c) Mass deposited

d) Fuel efficiency

96. Potentiometry determines:

a) Concentration of ions

b) Mass deposition

c) Temperature

d) Corrosion rate

97. Voltammetry is useful for:

a) Trace metal detection

b) Battery construction

c) Electrolysis only

d) Painting

98. Fuel cells convert:

a) Chemical energy → Electrical energy

b) Electrical energy → Chemical energy

c) Heat → Chemical energy

d) None

99. Electrochemistry helps in energy storage via:

a) Batteries and fuel cells

b) Electroplating

c) Molten salt electrolysis

d) Conductometry

100. Industrial application of electrochemistry:

a) Chlor-alkali process

b) Lead-acid battery

c) Electroplating

d) All of the above